

Module 15 – Agile Methodology

1. Introduction to Agile

What is Agile?

Agile is a software development methodology that focuses on iterative development, customer collaboration, continuous feedback, and frequent delivery of working software. It divides a project into small iterations called Sprints, allowing teams to adapt quickly to changing requirements.

Objectives of Agile

- Deliver working software quickly.
- Improve customer satisfaction.
- Adapt to changing requirements.
- Encourage team collaboration.
- Improve software quality.
- Reduce project risks.
- Ensure continuous improvement.

Benefits of Agile

- Faster software delivery.
- Better customer involvement.
- Higher product quality.
- Increased flexibility.
- Early bug detection.
- Continuous feedback.
- Improved team collaboration.
- Reduced project risk.

2. Agile Manifesto

The Agile Manifesto was introduced in **2001** by seventeen software developers.

Four Values of Agile Manifesto

1. Individuals and interactions over processes and tools.
2. Working software over comprehensive documentation.
3. Customer collaboration over contract negotiation.
4. Responding to change over following a plan.

Twelve Principles of Agile

1. Satisfy customers through early and continuous delivery.
2. Welcome changing requirements.
3. Deliver working software frequently.
4. Business people and developers work together daily.
5. Build projects around motivated individuals.
6. Face-to-face communication is the most effective.
7. Working software is the primary measure of progress.
8. Maintain a sustainable development pace.
9. Focus on technical excellence and good design.
10. Simplicity is essential.
11. Self-organizing teams produce the best designs.
12. Regularly reflect and improve team performance.

3. Agile vs Waterfall

Agile	Waterfall
Iterative development	Sequential development
Flexible to changes	Difficult to change requirements
Continuous customer feedback	Customer involved mainly at the beginning and end
Frequent software releases	Single release after completion
Faster delivery	Slower delivery
Lower project risk	Higher project risk

4. Scrum Framework

Scrum is the most widely used Agile framework for managing software development projects.

Scrum Roles

Product Owner

- Represents the customer.
- Defines product requirements.
- Prioritizes the Product Backlog.
- Maximizes product value.

Scrum Master

- Facilitates Scrum activities.
- Removes obstacles.
- Ensures Scrum practices are followed.
- Supports the development team.

Development Team

- Designs, develops, tests, and delivers the product.
- Self-organizing and cross-functional.
- Responsible for completing Sprint goals.

5. Scrum Ceremonies

Sprint Planning

The team selects Product Backlog items for the Sprint and defines Sprint goals.

Daily Scrum

A 15-minute daily meeting where team members discuss:

- What was completed yesterday?
- What will be done today?
- Are there any obstacles?

Sprint Review

Conducted at the end of the Sprint to demonstrate the completed product increment to stakeholders and collect feedback.

Sprint Retrospective

The team reviews the Sprint and discusses:

- What went well?
- What could be improved?
- Action items for the next Sprint.

6. Scrum Artifacts

Product Backlog

A prioritized list of all project requirements maintained by the Product Owner.

Sprint Backlog

A subset of Product Backlog items selected for the current Sprint.

Increment

The working software delivered at the end of each Sprint.

Definition of Done (DoD)

The Definition of Done specifies the conditions that must be satisfied before a Product Backlog item is considered complete.

Example:

- Code completed.
- Unit testing passed.
- Code reviewed.
- Documentation updated.
- Product Owner approval received.

7. Agile Estimation

Story Points

Story Points estimate the effort required to complete a User Story rather than the time required.

Common Story Point values:

- 1
- 2
- 3
- 5
- 8
- 13
- 21

Larger Story Points indicate greater complexity and effort.

Planning Poker

Planning Poker is an Agile estimation technique where team members independently estimate Story Points using numbered cards.

Steps:

1. Product Owner explains the User Story.
2. Team discusses the requirements.
3. Each member selects a Story Point card.
4. Cards are revealed simultaneously.
5. Differences are discussed.
6. Final estimate is agreed upon.

Benefits:

- Reduces estimation bias.
- Encourages team discussion.
- Improves estimation accuracy.

8. Sprint Planning

Sprint Planning is conducted at the beginning of every Sprint.

Activities include:

- Reviewing Product Backlog.
- Selecting User Stories.
- Estimating effort.
- Defining Sprint Goal.
- Creating Sprint Backlog.

9. Velocity

Velocity measures the amount of work completed by a Scrum team during one Sprint.

Formula:

Velocity = Total Story Points Completed in a Sprint

Example:

Sprint 1 = 25 Story Points

Sprint 2 = 30 Story Points

Sprint 3 = 28 Story Points

Average Velocity = $(25 + 30 + 28) / 3 = 27.67$ Story Points per Sprint

10. Burndown Chart

A Burndown Chart is a graphical representation showing the amount of work remaining during a Sprint.

Benefits:

- Tracks Sprint progress.
- Identifies delays.
- Helps predict Sprint completion.
- Improves Sprint planning.

11. Agile User Stories

What is a User Story?

A User Story is a simple description of a software feature written from the user's perspective.

Standard Format

As a *[user]*,

I want *[goal]*,

So that *[benefit]*.

Example:

As a customer, I want to reset my password so that I can regain access to my account if I forget it.

12. INVEST Principle

A good User Story should satisfy the **INVEST** principle.

Letter	Meaning
I	Independent
N	Negotiable
V	Valuable
E	Estimable

S	Small
T	Testable

13. Acceptance Criteria

Acceptance Criteria define the conditions that must be satisfied for a User Story to be accepted.

Given-When-Then Format

Given a specific condition,

When an action is performed,

Then an expected result should occur.

Example:

Given the user is on the login page,

When valid username and password are entered,

Then the user should be redirected to the dashboard.

14. Writing User Stories

Example 1

As a customer, I want to search for products so that I can quickly find the required items.

Example 2

As an administrator, I want to add new users so that employees can access the system.

Example 3

As a student, I want to view my examination results so that I can track my academic performance.

15. Advantages of Agile

- Faster software delivery.
- Better customer satisfaction.
- Continuous feedback.

- Improved collaboration.
- Better product quality.
- Easy adaptation to changing requirements.
- Reduced project risk.
- Continuous improvement.

16. Challenges of Agile

- Requires active customer involvement.
- Difficult to estimate project cost initially.
- Frequent requirement changes.
- Team members must be highly collaborative.
- Less documentation compared to traditional methods.

17. Conclusion

Agile is a flexible software development methodology that emphasizes collaboration, iterative development, customer feedback, and continuous improvement. Scrum provides a structured framework with defined roles, ceremonies, and artifacts, while techniques such as Story Points, Planning Poker, Velocity, Burndown Charts, and User Stories help teams estimate work, monitor progress, and deliver high-quality software incrementally.